

Chapter 12: Rates, Ratios and Time

Three-week revision booklet • 6 worksheets • 108 questions

NSW syllabus outcome

MA4-RAT-C-01 — solves problems involving ratios and rates •
MAO-WM-01

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How to use this booklet

There are six worksheets, two for each week, and each one runs across **two pages**. Every section opens with a short **worked example** showing you exactly how that kind of question is done. The numbers have been kept friendly: the Easy and Medium questions can mostly be done in your head, and the Challenging ones need only a few lines of written working. Each sheet is split into **Easy**, **Medium** and **Challenging**, and every question that needs working out is given ruled space for it. The final worksheet mixes every skill together, so it is the best check that you are ready for the topic test.

Week	Worksheet	Skill
1	1	Simplifying ratios
1	2	Equivalent ratios and dividing a ratio
2	3	Ratios with units and scale
2	4	Rates and best buy
3	5	Time and speed
3	6	Mixed revision

Progress tracker

Worksheet	Date completed	Score / 18	Checked by
1 Simplifying ratios			
2 Equivalent ratios and dividing a ratio			
3 Ratios with units and scale			
4 Rates and best buy			
5 Time and speed			
6 Mixed revision			

Worksheet 1: Simplifying ratios

Week 1 · Worksheet 1 of 6 · 21 questions

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A ratio compares two or more amounts. Simplify it by dividing every part by the same number.

EASY

12 QUESTIONS

SIMPLIFYING A RATIO

- Find a number that divides into **both** parts.
- $12 : 18$ — both divide by 6, so $12 : 18 = 2 : 3$.
- Keep going until no number divides both parts any more.

Simplify each ratio.

1. $4 : 6$

Answer: _____

2. $10 : 15$

Answer: _____

3. $8 : 12$

Answer: _____

4. $6 : 9$

Answer: _____

5. $20 : 30$

Answer: _____

6. $9 : 12$

Answer: _____

7. $5 : 10$

Answer: _____

8. $12 : 16$

Answer: _____

9. $14 : 21$

Answer: _____

10. $6 : 8$

Answer: _____

11. $15 : 20$

Answer: _____

12. $18 : 24$

Answer: _____

Worksheet 1: Simplifying ratios

Week 1 · Worksheet 1 of 6 · continued

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MEDIUM

6 QUESTIONS

THREE-PART RATIOS

- Exactly the same idea — the number you divide by must go into **every** part.
- $4 : 8 : 12$ — all divide by 4, so it becomes **$1 : 2 : 3$** .

Simplify each ratio.

13. $2 : 4 : 6$

Answer:

14. $5 : 10 : 15$

Answer:

15. $3 : 6 : 9$

Answer:

16. $4 : 8 : 12$

Answer:

17. $6 : 9 : 12$

Answer:

18. $10 : 20 : 30$

Answer:

CHALLENGING

3 QUESTIONS

RATIOS THAT ARE NOT WHOLE NUMBERS

- Multiply **both** parts by the same number first to clear decimals or fractions.
- $1.5 : 3$ — multiply both by 2 to get $3 : 6$, which is **$1 : 2$** .

Read each one, show your working, then answer.

19. A class has 12 boys and 18 girls. Write the ratio of boys to girls in its simplest form.

20. Simplify $1.5 : 3$. Multiply both parts by 2 first, then simplify.

21. Simplify $\frac{1}{2} : \frac{1}{4}$. Multiply both parts by 4 first, then simplify.

Draw your diagram here

Draw your diagram here

Draw your diagram here

Answer:

Answer:

Answer:

Worksheet 2: Equivalent ratios and dividing a ratio

Week 1 · Worksheet 2 of 6 · 18 questions

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Multiply or divide every part by the same number to find an equivalent ratio.

EASY

9 QUESTIONS

FINDING THE MISSING NUMBER

- Ask what the first part was multiplied by, then do the same to the other part.
- $1:3 = 4:? \rightarrow 1$ was multiplied by 4, so $3 \times 4 = 12$.

Complete each equivalent ratio.

1. $1:2 = 3: \underline{\hspace{1cm}}$

2. $2:3 = 4: \underline{\hspace{1cm}}$

3. $1:5 = 2: \underline{\hspace{1cm}}$

Answer: _____

Answer: _____

Answer: _____

4. $3:4 = 6: \underline{\hspace{1cm}}$

5. $2:5 = \underline{\hspace{1cm}}:10$

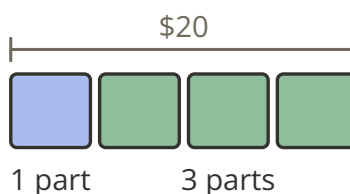
6. $1:3 = \underline{\hspace{1cm}}:9$

Answer: _____

Answer: _____

Answer: _____

\$20 is shared in the ratio 1 : 3. Use the bar model to help you.



7. How many parts are there altogether?

8. How much is one part worth?

9. How much is each share?

Answer: _____

Answer: _____

Answer: _____

Worksheet 2: Equivalent ratios and dividing a ratio

Week 1 · Worksheet 2 of 6 · continued

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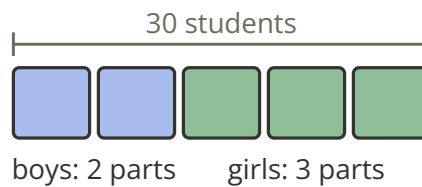
MEDIUM

6 QUESTIONS

SHARING IN A RATIO

- Add the parts to get the **total number of parts**.
- Share \$20 in 1 : 3 — that is 4 parts, so one part = $\$20 \div 4 = \5 .
- Then 1 part = \$5 and 3 parts = $3 \times \$5 = \15 . Check: $\$5 + \$15 = \$20$.

There are 30 students in the ratio 2 boys to 3 girls.



10. How many parts altogether?

Answer: _____

11. How many students is one part?

Answer: _____

12. How many boys and how many girls?

Answer: _____

Share each amount in the ratio given.

13. Share \$30 in the ratio 1 : 2

Answer: _____

14. Share 40 kg in the ratio 3 : 5

Answer: _____

15. Share 24 lollies in the ratio 1 : 2 : 3

Answer: _____

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Write down how many parts there are altogether, then find what one part is worth.

Read each one, show your working, then answer.

16. Two friends share \$50 in the ratio 2 : 3. How much does each friend get?

Draw your diagram here

Answer: _____

17. A recipe uses flour and sugar in the ratio 3 : 1. If 300 g of flour is used, how much sugar is needed?

Draw your diagram here

Answer: _____

18. The ratio of boys to girls in a class is 2 : 3. There are 12 boys. How many girls are there, and how many students altogether?

Draw your diagram here

Answer: _____

Worksheet 3: Ratios with units and scale

Week 2 · Worksheet 3 of 6 · 21 questions

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Change both amounts into the same unit first, then simplify.

EASY

12 QUESTIONS

MAKING THE UNITS MATCH

- Change the larger unit into the smaller one first.
- 1 m : 50 cm — 1 m is 100 cm, so it is $100 : 50 = 2 : 1$.
- Useful facts: 1 m = 100 cm, 1 km = 1000 m, 1 kg = 1000 g, 1 h = 60 min.

Simplify each ratio. Change the units to match first.

1. 1 m : 50 cm

2. 2 km : 500 m

3. 1 h : 30 min

Answer: _____

Answer: _____

Answer: _____

4. 1 kg : 250 g

5. 1 L : 500 mL

6. 2 m : 40 cm

Answer: _____

Answer: _____

Answer: _____

7. 1 h : 15 min

8. 1 km : 250 m

9. 3 kg : 500 g

Answer: _____

Answer: _____

Answer: _____

10. 1 km : 500 m

11. 2 kg : 400 g

12. 1 m : 25 cm

Answer: _____

Answer: _____

Answer: _____

Worksheet 3: Ratios with units and scale

Week 2 · Worksheet 3 of 6 · continued

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MEDIUM

6 QUESTIONS

MONEY AND TIME

- Change dollars into cents: \$2 : 50c is 200 : 50 = **4 : 1**.
- Change days into hours, or weeks into days, before you simplify.

Simplify each ratio.

13. \$2 : 50c

14. 1 day : 6 hours

15. 2 L : 250 mL

Answer:

Answer:

Answer:

16. 1 week : 2 days

17. 4 m : 80 cm

18. 1.5 m : 50 cm

Answer:

Answer:

Answer:

CHALLENGING

3 QUESTIONS

USING A SCALE

- A scale of 1 cm : 100 m means every 1 cm on the map is 100 m in real life.
- So 3 cm on the map is $3 \times 100 = 300$ m in real life.

A map has a scale of 1 cm : 100 m. Show your working.

19. Two towns are 4 cm apart on the map. How far apart are they in real life?

20. A road is 600 m long in real life. How long is it on the map?

21. A model car is built to a scale of 1 : 20. The model is 15 cm long. How long is the real car, in centimetres and then in metres?

Draw your diagram here

Draw your diagram here

Draw your diagram here

Answer:

Answer:

Answer:

Worksheet 4: Rates and best buy

Week 2 · Worksheet 4 of 6 · 21 questions

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A rate compares two amounts with different units. A unit rate is the amount for just one.

EASY

12 QUESTIONS

FINDING A UNIT RATE

- Divide by the number of items, kilograms or hours.
- \$12 for 4 kg — $12 \div 4 =$ **\$3 per kg.**
- 100 km in 2 hours — $100 \div 2 =$ **50 km/h.**

Find the unit rate.

1. \$12 for 4 kg

2. 100 km in 2 hours

3. \$20 for 5 items

Answer: _____

Answer: _____

Answer: _____

4. 60 words in 2 minutes

5. \$9 for 3 books

6. 150 km in 3 hours

Answer: _____

Answer: _____

Answer: _____

7. \$30 for 6 pens

8. 240 km in 4 hours

9. \$10 for 5 apples

Answer: _____

Answer: _____

Answer: _____

10. \$16 for 8 kg

11. 200 km in 4 hours

12. \$15 for 3 tickets

Answer: _____

Answer: _____

Answer: _____

Worksheet 4: Rates and best buy

Week 2 · Worksheet 4 of 6 · continued

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MEDIUM

6 QUESTIONS

FINDING THE BETTER BUY

- Work out the price for **one** of each, then compare.
- 500 g for \$4 is \$8 per kg; 1 kg for \$7 is \$7 per kg, so the **1 kg pack** is better.
- The cheaper unit price is always the better buy.

Work out the price of one, then say which is the better buy.

13. 500 g for \$4, or 1 kg for \$7

14. 4 pens for \$8, or 6 pens for \$9

15. 2 L for \$6, or 5 L for \$12.50

Answer:

Answer:

Answer:

16. 3 kg for \$12, or 5 kg for \$18

17. 250 mL for \$2, or 1 L for \$7

18. 6 eggs for \$3, or 12 eggs for \$5.40

Answer:

Answer:

Answer:

CHALLENGING

3 QUESTIONS

USING A RATE

- Multiply the rate to find the total, or divide by the rate to find how many.
- At \$8 per kg, 5 kg costs $5 \times \$8 = \40 .

Read each one, show your working, then answer.

19. Mince costs \$8 per kilogram. How much does 5 kg cost?

20. Petrol costs \$2 per litre. How many litres can be bought with \$50?

21. A car travels 180 km in 3 hours. Find its speed, then work out how far it would travel in 5 hours at the same speed.

Draw your diagram here

Draw your diagram here

Draw your diagram here

Answer:

Answer:

Answer:

Worksheet 5: Time and speed

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Week 3 · Worksheet 5 of 6 · 18 questions

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In 24-hour time the afternoon hours run from 13:00 to 23:59. Speed = distance ÷ time.

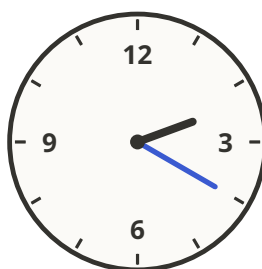
EASY

9 QUESTIONS

CHANGING BETWEEN 12-HOUR AND 24-HOUR TIME

- Before noon: keep the hour and write it with two digits. 7:05 a.m. = **07:05**.
- After noon: **add 12** to the hour. 3:15 p.m. = **15:15**.
- Going back: take 12 off any hour bigger than 12. 20:40 = **8:40 p.m.**

The clock shows an afternoon time.



What time is this?

1. Write the time in 12-hour time.

2. Write the time in 24-hour time.

3. What time will it be 30 minutes later?

Answer: _____

Answer: _____

Answer: _____

Convert to 24-hour time.

4. 3:15 p.m.

5. 7:05 a.m.

6. 9:30 p.m.

Answer: _____

Answer: _____

Answer: _____

Convert to 12-hour time.

7. 14:20

8. 08:45

9. 19:00

Answer: _____

Answer: _____

Answer: _____

Worksheet 5: Time and speed

Week 3 • Worksheet 5 of 6 • continued

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MEDIUM

6 QUESTIONS

TIME DIFFERENCES AND SPEED

- Count on in whole hours first, then the minutes.
- From 9:20 a.m. to 11:50 a.m. is 2 hours, then 30 minutes = **2 h 30 min**.
- Speed = distance $\div$ time. 100 km in 2 h = **50 km/h**.

Find each time difference.

10. 9:20 a.m. to 11:50 a.m.

Answer:

11. 7:45 a.m. to 10:15 a.m.

Answer:

12. A film starts at 18:30 and runs for 90 minutes. When does it end?

Answer:

Answer each question.

13. A car travels 100 km in 2 hours. Find its speed.

Answer:

14. A truck travels for 3 hours at 60 km/h. Find the distance.

Answer:

15. How long does it take to travel 120 km at 40 km/h?

Answer:

CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Write the times in 24-hour form first — it makes the subtraction easier.

Read each one, show your working, then answer.

16. A train leaves at 09:15 and arrives at 12:45. How long is the journey?

Draw your diagram here

Answer:

17. A cyclist rides 30 km in 2 hours. Find the speed, then work out how far the cyclist would ride in 5 hours at that speed.

Draw your diagram here

Answer:

18. A bus leaves at 2:30 p.m. and the trip takes 1 hour 45 minutes. What time does it arrive? Give your answer in 24-hour time.

Draw your diagram here

Answer:

Worksheet 6: Mixed revision

NAME _____

Week 3 · Worksheet 6 of 6 · 21 questions

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This sheet mixes every skill from the Rates, Ratios and Time chapter. Work through it as though it were the topic test.

EASY

12 QUESTIONS

QUICK REMINDERS

- Simplify a ratio by dividing every part by the same number.
- Match the units before you simplify: 1 m : 50 cm = 100 : 50 = **2 : 1**.
- A unit rate is the amount for one: \$12 for 4 kg = **\$3 per kg**.

Simplify each ratio.

1. 4 : 6

2. 10 : 15

3. 2 : 4 : 6

Answer: _____

Answer: _____

Answer: _____

Simplify each ratio. Change the units to match first.

4. 1 m : 50 cm

5. 1 h : 30 min

6. 1 kg : 250 g

Answer: _____

Answer: _____

Answer: _____

Find the unit rate.

7. \$12 for 4 kg

8. 100 km in 2 hours

9. \$20 for 5 items

Answer: _____

Answer: _____

Answer: _____

Convert to 24-hour time.

10. 3:15 p.m.

11. 7:05 a.m.

12. 9:30 p.m.

Answer: _____

Answer: _____

Answer: _____

Worksheet 6: Mixed revision

Week 3 • Worksheet 6 of 6 • continued

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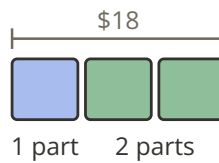
● MEDIUM

6 QUESTIONS

CHOOSING WHAT TO DO

- Sharing? Add the parts, then find what one part is worth.
- Better buy? Work out the price of one of each and compare.
- Speed = distance $\div$ time.

\$18 is shared in the ratio 1 : 2. Use the bar model to help you.



13. How much is one part worth?

Answer: _____

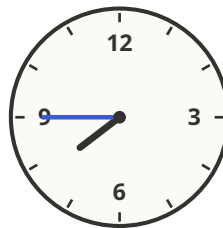
14. How much is each share?

Answer: _____

15. Check that the two shares add to \$18.

Answer: _____

The clock shows a morning time.



What time is this?

16. Write the time in 24-hour time.

Answer: _____

17. What time will it be 20 minutes later?

Answer: _____

18. How long is it until 10:00 a.m.?

Answer: _____

● CHALLENGING

3 QUESTIONS

BEFORE YOU START

- Write down what you know first, then decide which method the question needs.

Read each one, show your working, then answer.

19. Two friends share \$50 in the ratio 2 : 3. How much does each get?

20. Which is the better buy: 500 g for \$4, or 1 kg for \$7? Work out the price per kilogram for each.

21. A car travels 90 km in 2 hours. Find its speed, then work out how far it would travel in 5 hours at the same speed.

Draw your diagram here

Draw your diagram here

Draw your diagram here

Answer: _____

Answer: _____

Answer: _____